

Resilient Trust–Aware Distributed Observer Design for Connected Vehicle Platoons

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Abstract: This paper proposes a trust-aware distributed observer for resilient state estimation in vehicle platoons under cyberattacks. A behavioral divergence metric scores shared data and builds a dynamic trusted-neighbor set for online weight adaptation. Sufficient ISS stability conditions are provided. Simulations under bogus, replay, and DoS attacks show improved estimation resilience.

Keywords: Distributed observers, secure estimation, Lyapunov stability, cyber-attack.

1. INTRODUCTION AND VEHICLE MODELING

1.1 General introduction

Connected vehicle platoons use vehicle-to-vehicle (V2V) communication and distributed observers to reconstruct platoon states from onboard and neighboring data. This cooperation improves safety and traffic efficiency, but faulty or malicious agents can send bogus or replayed data, or drop packets, causing estimation errors to spread through the network (Abbaszadeh and Zemouche, 2022; Meng et al., 2025).

Most distributed observers assume reliable data exchange. Resilient alternatives use residual tests, consensus cross-checking, or trust scores Nguyen et al. (2024); Barboni et al. (2019); Yang et al. (2022); Yemini et al. (2022); Wang et al. (2021). These methods improve detection, but fallback modes or fixed trust weights may react slowly to time-varying attacks Mustafa et al. (2020); Li et al. (2024).

This paper proposes a trust-aware distributed observer that compares reported and predicted behavior, updates neighbor trust online, and adapts observer weights. The contributions are: a resilient estimation formulation under adversarial communication; a behavioral trust score for trusted-neighbor selection; and a trust-weighted observer with ISS stability conditions, validated under bogus, replay, and DoS attacks.

1.2 Vehicle mathematical model

The platoon is represented by a weighted communication graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{A}\}$, where $\mathcal{V} = \{1, \dots, N\}$, $\mathcal{A} = [a_{ij}]$, and $\mathcal{N}_i = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}\}$ denotes the neighbor set of vehicle i . The Laplacian $\mathcal{L} = [l_{ij}]$ is defined by $l_{ii} = \sum_{k=1}^N a_{ik}$ and $l_{ij} = -a_{ij}$ for $i \neq j$. The discrete-time longitudinal dynamics of each vehicle $i \in \mathcal{V}$ are

$$\begin{aligned} x_i(t+1) &= A_b x_i(t) + B_b u_i(t) + \Delta_i(t), \\ y_i(t) &= C_i x_i(t), \end{aligned} \quad (1)$$

where $x_i = [s_i \ v_i \ a_i]^\top$ contains position, velocity, and acceleration, and u_i is the control input. The system matrices are

$$A_b = \begin{bmatrix} 1 & T_s & \frac{T_s^2}{2} \\ 0 & 1 & T_s \\ 0 & 0 & 1 - \frac{T_s}{\tau_b} \end{bmatrix}, \quad B_b = \begin{bmatrix} 0 \\ 0 \\ \frac{T_s}{\tau_b} \end{bmatrix}, \quad C_b = I_3,$$

where T_s is the sampling time, $\tau_b > 0$ is the nominal engine lag, and $\Delta_i(t)$ captures modeling uncertainties.

With the collective state $x = \text{col}(x_1, \dots, x_N)$, the platoon dynamics are

$$\begin{cases} x(t+1) = (I_N \otimes A_b)x(t) + (I_N \otimes B_b)u(t) + \Delta(t) \\ y_i(t) = C_i x_i(t), \end{cases} \quad (2)$$

where $u = \text{col}(u_1, \dots, u_N)$, $\Delta = \text{col}(\Delta_1, \dots, \Delta_N)$, and $C_i = \begin{bmatrix} 0 & \dots & \underbrace{C_b}_{i\text{th}} & \dots & 0 \end{bmatrix}$. Each vehicle measures its own full state, so (A_b, C_b) is observable. However, $((I_N \otimes A_b), C_i)$ is unobservable for the full platoon dynamics (2), which motivates distributed estimation through inter-vehicle communication.

2. OBSERVER ARCHITECTURE

The distributed observer estimates the full platoon state x . This section introduces the observer architecture; the trust mechanism is detailed in Section 3. Two layers are used: a Local Observer ($\mathcal{LO}$) based on onboard sensors, and a Distributed Observer ($\mathcal{DO}$) that fuses information exchanged through communication links.

2.1 General structure of the observer

The observer is defined by

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$$(\mathcal{LO}_i) : \hat{x}_0^{(i)}(t+1) = A_b \hat{x}_0^{(i)}(t) + B_b u_i(t) + F_i (y_i(t) - C_b \hat{x}_0^{(i)}(t)) \quad (3a)$$

$$(\mathcal{DO}_i^{(j)}) : \hat{x}_i^{(j)}(t+1) = A_b \hat{x}_i^{(j)}(t) + B_b \hat{u}_j(t) + A_b \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t) (\hat{x}_l^{(j)}(t) - \hat{x}_i^{(j)}(t)) + A_b w_{i0}^{(j)}(t) (\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t)) \quad (3b)$$

where $\hat{x}_0^{(i)}(t)$ is the local-observer state of vehicle i , F_i is the corresponding gain, and $\mathcal{N}_i$ is the neighbor set of vehicle i in $\mathcal{G}$. The variable $\hat{x}_i^{(j)}(t)$ is vehicle i 's distributed estimate of vehicle j 's state. The weights $w_{il}^{(j)}(t)$, $l \in \mathcal{N}_i$, and $w_{i0}^{(j)}(t)$ scale the neighbor-consensus and local-observer anchoring terms. In the nominal case, the desired objectives are

$$\lim_{t \rightarrow \infty} (\hat{x}_0^{(i)}(t) - x_i(t)) = 0, \quad (4)$$

$$\lim_{t \rightarrow \infty} (\hat{x}_i(t) - x(t)) = 0, \quad (5)$$

where $\hat{x}_i(t) = \text{col}(\hat{x}_i^{(1)}(t), \dots, \hat{x}_i^{(N)}(t))$ is the collective distributed estimate maintained by vehicle i .

Remark 1. If a host forms $\hat{u}_j(t)$ (e.g., from a known controller law or an input estimator), the mismatch

$$\Delta u_j(t) = u_j(t) - \hat{u}_j(t) \quad (6)$$

will be treated as part of the disturbance vector. We do not require $\hat{u}_j(t)$ for the $(\mathcal{DO})$ to run; using it only tightens bounds.

2.2 On the structure of the observer (3)

The distributed observer $\mathcal{DO}_i^{(j)}$ estimates the state of vehicle $j \in \mathcal{V}$ from vehicle i . The neighbor-consensus term in (3b) aligns $\hat{x}_i^{(j)}(t)$ with estimates received from $l \in \mathcal{N}_i$, while $w_{il}^{(j)}(t)$ sets each neighbor's influence. Since consensus alone may drive all estimates to a common biased value under disturbances or model mismatch, the anchoring term $w_{i0}^{(j)}(t)(\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t))$ injects the local estimate of vehicle j as a reference.

2.3 Introduction of a virtual communication graph

Each distributed observer $\mathcal{DO}_i^{(j)}$ needs reference information from the local observer $\mathcal{LO}_j$. To include this reference, we define a virtual node "0" connected to vehicle j and to its neighbors $\mathcal{N}_j$ in the original graph $\mathcal{G}$. The virtual graph $\bar{\mathcal{G}}^{(j)} = \{\bar{\mathcal{V}}, \bar{\mathcal{E}}^{(j)}\}$ is

$$\bar{\mathcal{V}} = \mathcal{V} \cup \{0\}, \quad (7)$$

$$\bar{\mathcal{E}}^{(j)} = \mathcal{E} \cup \{(0, j)\} \cup \{(0, k) | k \in \mathcal{N}_j\}.$$

The corresponding adjacency matrix $\bar{\mathcal{A}}^{(j)} \in \mathbb{R}^{(N+1) \times (N+1)}$ is

$$\bar{\mathcal{A}}^{(j)} = \begin{bmatrix} 0 & 0 \\ w_0^{(j)} & \mathcal{A} \end{bmatrix}, \quad (8)$$

where the vector $w_0^{(j)} = \text{col}(w_{10}^{(j)}, \dots, w_{N0}^{(j)}) \in \mathbb{R}^N$ defines the connection strengths from node 0 to the distributed observers.

This graph combines local estimation through node 0 and neighbor consensus through $\mathcal{G}$, so each local observer can act as an anchor for distributed estimation.

2.4 Main objectives

Fixed observer weights assume reliable data exchange, which is not valid when compromised vehicles transmit falsified information. The objective is to define a trust measure for received data and use it to update $w_{il}^{(j)}(t)$ and $w_{i0}^{(j)}(t)$ online, while choosing F_i so the synthesis in Section 4 yields an ISS convergence bound.

3. TRUST SCORE FRAMEWORK

To support resilient estimation, each vehicle evaluates the reliability of received data. The proposed trust framework quantifies how much a host vehicle i trusts another vehicle l . Its structure is shown in Fig. 1.

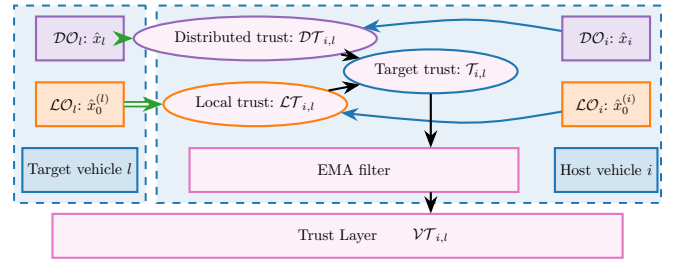


Fig. 1. Trust framework.

In this context, we define the following types of trust:

- **Local Trust** $\mathcal{LT}_{i,l}$: reliability of vehicle l 's local-observer data.
- **Distributed Trust** $\mathcal{DT}_{i,l}$: agreement between distributed estimates from l and those of i .
- **Target Trust** $\mathcal{T}_{i,l}$: instantaneous fusion of Local Trust and Distributed Trust.
- **Vehicle Trust** $\mathcal{VT}_{i,l}$: EMA-smoothed trust score used by the observer.

The following subsections define these scores and their use in the observer.

3.1 Data validity indicator

Before trust computation, each received V2V packet is checked for authenticity and freshness:

$$\delta_{i,l}(t) = \begin{cases} 1, & \text{if the message is authenticated and fresh,} \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Only packets with $\delta_{i,l}(t) = 1$ are used for trust evaluation.

3.2 Local trust

Local Trust $\mathcal{LT}_{i,l}$ measures the confidence of host vehicle i in the local-observer output of neighbor $l \in \mathcal{N}_i$. It uses three consistency indicators: velocity, distance, and acceleration. Missing data are handled by a hold-and-decay rule.

Velocity consistency: The host compares its local velocity estimate with the velocity reported by neighbor l :

$$\iota_{i,l}^{\text{velocity}} = \max \left(1 - \frac{|\hat{v}_0^{(l)} - \hat{v}_0^{(i)}|}{\varepsilon_v + |\hat{v}_0^{(i)}|}, 0 \right)$$

where $\varepsilon_v > 0$ avoids division by zero. This score uses only neighbor-to-host velocity consistency and does not require leader information.

Distance consistency: The host vehicle i computes the signed distance to target l as

$$d_{i,l} = \hat{s}_0^{(l)} - \hat{s}_0^{(i)} - L_l,$$

where L_l is the length of vehicle l . The mismatch with the onboard measured distance $d_{i,l}^{\text{meas}}$ is

$$\iota_{i,l}^{\text{distance}} = \max \left(1 - \frac{|d_{i,l} - d_{i,l}^{\text{meas}}|}{\varepsilon_d + |d_{i,l}^{\text{meas}}|}, 0 \right)$$

where $\varepsilon_d > 0$ avoids division by zero.

Assumption 1. When a vehicle is connected, it has access to basic information about its neighbors, including their type and length.

Acceleration consistency: The host estimates relative acceleration as

$$a_{\text{recv}}^{\text{rel}} = \hat{a}_0^{(l)} - \hat{a}_0^{(i)}, \quad a_{\text{expect}}^{\text{rel}} = \frac{v_{\text{rel}}(t) - v_{\text{rel}}(t-n)}{nT_s},$$

where $v_{\text{rel}} = \hat{v}_0^{(l)} - \hat{v}_0^{(i)}$ and n is the averaging window. The mismatch is normalized by distance and relative velocity:

$$\iota_{i,l}^{\text{accel}} = \max \left(1 - \frac{|a_{\text{recv}}^{\text{rel}} - a_{\text{expect}}^{\text{rel}}|}{\varepsilon_a + a_{\text{th}} \left(1 + \frac{|d_{i,l}|}{d_{\text{norm}}} \right) \left(1 + \frac{|v_{\text{rel}}|}{v_{\text{norm}}} \right)}, 0 \right)$$

where $\varepsilon_a > 0$ avoids division by zero, $a_{\text{th}} > 0$ is a nominal acceleration tolerance, and $d_{\text{norm}} > 0$ and $v_{\text{norm}} > 0$ are normalization constants.

This normalization penalizes inconsistencies more when vehicles are close or have large relative speed, while reducing false trust drops from minor sensor fluctuations.

Fusion and update rule: The three indicators are fused into the local trust score:

$$\mathcal{LT}_{i,l}(t) = \begin{cases} f(\iota_{i,l}^{\text{velocity}}, \iota_{i,l}^{\text{distance}}, \iota_{i,l}^{\text{accel}}) & \text{if } \delta_{i,l}(t) = 1 \\ (1 - \lambda_h) \mathcal{LT}_{i,l}(t-1), & \text{otherwise} \end{cases} \quad (10)$$

where f combines the three scores and λ_h sets the decay rate during data loss.

Remark 2. Function f can be a weighted average or a minimum, depending on the desired sensitivity. The decay λ_h can increase with consecutive missed packets.

3.3 Distributed trust

Distributed Trust applies the same idea to exchanged distributed estimates. It is defined as

$$\mathcal{DT}_{i,l}(t) = \begin{cases} \gamma_{i,l}^{\text{self}}(t) \cdot \gamma_{i,l}^{\text{local}}(t) \cdot \gamma_{i,l}^{\text{host}}(t) & \text{if } \delta_{i,l}(t) = 1 \\ (1 - \lambda_h) \mathcal{DT}_{i,l}(t-1), & \text{otherwise} \end{cases} \quad (11)$$

where $\gamma_{i,l}^{\text{self}}$, $\gamma_{i,l}^{\text{host}}$, and $\gamma_{i,l}^{\text{local}}$ check consistency with the sender's local observer, the host's distributed estimate, and the host's local measurements, respectively.

Sender self-consistency: The sender's distributed estimate of its own state should remain close to its local observer state:

$$\gamma_{i,l}^{\text{self}}(t) = \exp \left(- (\hat{x}_l^{(l)}(t) - \hat{x}_0^{(l)}(t))^{\top} \Sigma_{\text{self}}^{-1} \times (\hat{x}_l^{(l)}(t) - \hat{x}_0^{(l)}(t)) \right), \quad (12)$$

where Σ_{self} is the covariance matrix associated with the sender self-consistency residual.

Host-estimate consistency: The host compares the estimate received from target l with its own estimate using the Mahalanobis distance:

$$\rho_{i,l}^{(j)}(t) = \left(\hat{x}_i^{(j)}(t) - \hat{x}_l^{(j)}(t) \right)^{\top} \Sigma_{\text{host}}^{-1} \left(\hat{x}_i^{(j)}(t) - \hat{x}_l^{(j)}(t) \right) \quad (13)$$

where Σ_{host} is the diagonal covariance matrix of state-channel discrepancies:

$$\Sigma_{\text{host}} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2), \quad (14)$$

where n is the number of states and σ_k^2 is the variance of the k th state discrepancy.

The total discrepancy score is

$$D_{i,l}^{\text{host}}(t) = \sum_{j=1}^N \rho_{i,l}^{(j)}(t). \quad (15)$$

A small $D_{i,l}^{\text{host}}(t)$ means that vehicle l 's estimates are close to the host's estimates. The corresponding trust factor is

$$\gamma_{i,l}^{\text{host}}(t) = \exp(-D_{i,l}^{\text{host}}(t)). \quad (16)$$

where $\gamma_{i,l}^{\text{host}}(t) \in (0, 1]$.

Consistency with host's local measurement: Host vehicle i uses relative measurements to nearby vehicles, e.g., distance and relative velocity, to check whether the estimates from target l are consistent with local sensing.

Let $y_{i,j}(t) \in \mathbb{R}^{m_{ij}}$ be the relative measurement from vehicle i to vehicle j , e.g., $y_{i,j}(t) = s_j(t) - s_i(t)$ for relative position.

The estimate received from target l should approximately satisfy

$$H(\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t)) \approx y_{i,j}(t).$$

where $H \in \mathbb{R}^{m_{ij} \times n}$ extracts the measured states.

The consistency error is

$$\varepsilon_{i,l}^{(j)}(t) = H(\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t)) - y_{i,j}(t), \quad (17)$$

Using Σ_{local} as in (13), define

$$E_{i,l}^{(j)}(t) = (\varepsilon_{i,l}^{(j)}(t))^{\top} \Sigma_{\text{local}}^{-1} \varepsilon_{i,l}^{(j)}(t). \quad (18)$$

The total consistency error is

$$E_{i,l}(t) = \sum_{j \in \mathcal{N}_i} E_{i,l}^{(j)}(t). \quad (19)$$

A higher $E_{i,l}(t)$ means that the estimates from target l disagree with local relative measurements.

The trust factor is

$$\gamma_{i,l}^{\text{local}}(t) = \exp(-E_{i,l}(t)), \quad (20)$$

where $\gamma_{i,l}^{\text{local}}(t) \in (0, 1]$.

Remark 3. The covariance matrices Σ_{host} and Σ_{local} can be estimated from a recent nominal-data window at vehicle i , avoiding covariance transmission between nodes.

3.4 Target trust and temporal evolution

We combine the Local Trust and Distributed Trust into a single instantaneous trust score for target l :

$$\mathcal{T}_{i,l}(t) = \mathcal{L}\mathcal{T}_{i,l}(t) \cdot \mathcal{D}\mathcal{T}_{i,l}(t). \quad (21)$$

$\mathcal{T}_{i,l}(t)$ measures the instantaneous quality of target l 's data. Because this value can vary abruptly, a temporal filter is used.

$$\mathcal{V}\mathcal{T}_{i,l}(t) = (1 - \beta_{\text{trust}})\mathcal{V}\mathcal{T}_{i,l}(t-1) + \beta_{\text{trust}}\mathcal{T}_{i,l}(t), \quad (22)$$

where $\beta_{\text{trust}} \in [0, 1]$ is the EMA gain. A larger value reacts faster to attacks, while a smaller value gives smoother trust evolution. The observer uses $\mathcal{V}\mathcal{T}_{i,l}(t)$ to weigh information received from vehicle l .

4. TRUST-BASED OBSERVER WEIGHT DESIGN

The trust framework provides each vehicle with smoothed vehicle-trust scores $\mathcal{V}\mathcal{T}_{i,l}(t)$ representing the reliability of information from other vehicles. This section shows how these values are used to define adaptive observer weights $w_{il}^{(j)}(t)$ and establishes conditions for stability of the estimation error dynamics.

4.1 Weight based on the trust

Using the virtual graph $\bar{\mathcal{G}}^{(j)}$, each vehicle i selects its trusted neighbors at time t :

$$\begin{aligned} \mathcal{L}\mathcal{N}_i^{(j)}(t) = & \left\{ l \in \mathcal{N}_i \mid \mathcal{V}\mathcal{T}_{i,l}(t) \geq \theta_{\min} \right\} \\ & \cup \begin{cases} \{0\}, & \text{if } \mathcal{L}\mathcal{T}_{i,j}(t) \geq \theta_{\min}, \\ \emptyset, & \text{otherwise.} \end{cases} \end{aligned} \quad (23)$$

To limit the influence of each neighbor, the weights are normalized by

$$n_{il}^{(j)}(t) = \max \left\{ \kappa, \left| \mathcal{L}\mathcal{N}_i^{(j)}(t) \right| + 1 \right\} \geq 1, \quad \forall l \in \mathcal{L}\mathcal{N}_i^{(j)}(t) \quad (24)$$

where $\kappa \geq 1$ limits the maximum neighbor influence. The observer weights are

$$w_{il}^{(j)}(t) = \begin{cases} \frac{1}{n_{il}^{(j)}(t)}, & l \in \mathcal{L}\mathcal{N}_i^{(j)}(t) \\ 0, & \text{otherwise.} \end{cases} \quad (25)$$

These weights induce a switching communication graph. We assume:

Assumption 2. (*Jointly connected graph*). There exists a scalar constant $T > 0$ such that for all $t \geq 0$, the union graph $\bigcup_{\tau \in [t, t+T]} \bar{\mathcal{G}}^{(j)}(\tau)$ contains a spanning tree.

Assumption 2 means that every node $i \in \mathcal{V}$ is reachable from node “0” over each time window $[t, t+T]$, ensuring enough connectivity for the observer design Hao and Liu (2024).

4.2 Stability of the error

We now state sufficient conditions for stability of the estimation error dynamics. The errors satisfy

$$e_0^{(j)}(t+1) = (A_b - F_j C_b) e_0^{(j)}(t) + d_0^{(j)}(t), \quad (26)$$

$$\begin{aligned} e_{\mathcal{DO}}^{(j)}(t+1) = & \left(w_0^{(j)}(t) \otimes A_b \right) e_0^{(j)}(t) \\ & + \left((I_N - \bar{\mathcal{L}}^{(j)}(t)) \otimes A_b \right) e_{\mathcal{DO}}^{(j)}(t) + d_{\mathcal{DO}}^{(j)}(t), \end{aligned} \quad (27)$$

with

$$\bar{\mathcal{L}}^{(j)}(t) = \left[\bar{l}_{ik}^{(j)}(t) \right], \bar{l}_{ii}^{(j)}(t) = \sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t),$$

$$\bar{l}_{ik}^{(j)}(t) = \begin{cases} -w_{ik}^{(j)}(t), & k \neq i, k \in \mathcal{N}_i, \\ 0, & k \neq i, k \notin \mathcal{N}_i. \end{cases}$$

$$d_0^{(j)}(t) = -\Delta_j(t),$$

$$d_{\mathcal{DO}}^{(j)}(t) = (I_N \otimes B_b)(\hat{u}_j(t) - u_j(t)) - (\mathbf{1}_N \otimes \Delta_j(t)),$$

$$e_{\mathcal{DO}}^{(j)}(t) = \text{col} \left(e_1^{(j)}(t), \dots, e_N^{(j)}(t) \right)$$

where $w_0^{(j)}(t) = \text{col} \left(w_{10}^{(j)}(t), \dots, w_{N0}^{(j)}(t) \right)$ is a vector describing the connection between the local observer ($\mathcal{LO}_j$) and other distributed observers ($\mathcal{DO}_i$), $i \in \mathcal{V}$; $\bar{\mathcal{L}}^{(j)}(t)$ is the virtual weighted Laplacian matrix of the virtual graph $\bar{\mathcal{G}}^{(j)}(t)$.

We assume bounded disturbances.

Assumption 3. The disturbance vectors are bounded: there exist $\bar{d}_0 \geq 0$ and $\bar{d}_{\mathcal{DO}} \geq 0$ such that, for all $t \geq 0$,

$$\|d_0^{(j)}(t)\| \leq \bar{d}_0, \quad \|d_{\mathcal{DO}}^{(j)}(t)\| \leq \bar{d}_{\mathcal{DO}}.$$

The main ISS result is stated as follows.

Theorem 1. Consider a vehicle platoon (2) over a directed weighted communication network $\bar{\mathcal{G}}^{(j)}$. Let (3) be the corresponding distributed observer. Assume that the following conditions hold true:

- (1) For all $j \in \mathcal{V}$, there exists F_j such that the matrix $(A_b - F_j C_b)$ is *Schur* stable.
- (2) For all $i \in \mathcal{V}$, $w_{i0}^{(j)} \geq 0$ and $w_{il}^{(j)} \geq 0$ satisfy

$$\sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t) \leq 1. \quad (28)$$

- (3) There exists $\alpha \in (0, 1)$ and a matrix norm $\|\cdot\|_*$ such that, uniformly over all admissible switches,

$$\left\| (I_N - \bar{\mathcal{L}}^{(j)}(t)) \otimes A_b \right\|_* \leq \alpha < 1. \quad (29)$$

Then, the estimation error (26)–(27) is ISS. In particular, there exist constants $c_0, c_1 > 0$ and $\lambda \in (0, 1)$ such that

$$\|e_0^{(j)}(t)\| \leq \lambda^t \|e_0^{(j)}(0)\| + c_0 \bar{d}_0, \quad (30)$$

$$\begin{aligned} \|e_{\mathcal{DO}}^{(j)}(t)\| \leq & \lambda^t \|e_{\mathcal{DO}}^{(j)}(0)\| \\ & + c_1 \left(\|e_0^{(j)}(0)\| + \bar{d}_0 + \bar{d}_{\mathcal{DO}} \right). \end{aligned} \quad (31)$$

Remark 4. The conditions of Theorem 1 can be checked with standard Lyapunov arguments. The cascade structure allows the local error (26) and distributed error (27) to be analyzed separately.

5. SIMULATION RESULTS

This section evaluates estimation accuracy and trust evolution under falsified, faulty, and DoS communication attacks.

5.1 Simulation setup

The simulation uses four vehicles with all-to-all V2V communication. The observer runs onboard each vehicle, and small modeling variations are treated as uncertainties.

The main parameters are $N = 4$, $T_s = 0.1$ s, $\tau_b = 1.5$ s, local-trust weights $(w_v, w_d, w_a) = (1.2, 2.0, 0.9)$, EMA gain $\beta_{\text{trust}} = 0.5$, $d_{\text{norm}} = 15$ m, and $\theta_{\min} = 0.5$.

5.2 Attack scenarios

Six attacks on the leader data are considered, as summarized in Table 1. Each attack is broadcast to all followers during $[10, 15]$ s, and p denotes the attack or fault occurrence probability.

Table 1. Six attack cases in the scenario.

Case	Type	Target	Parameters
1	Bias	Position	bias = -5 m
2	Faulty	Position	int. = 10, $p = 0.3$
3	Bias	Velocity	bias = -2 m/s
4	Faulty	Velocity	int. = 2.5, $p = 0.3$
5	Faulty	Acceleration	int. = 1.0, $p = 0.3$
6	Drop	All	$p_{\text{drop}} = 0.5$

5.3 Results

Figure 2 reports the normalized impact score, defined as the raw combined estimation error divided by the maximum error over all vehicles and attacks. The DoS attack (Case 6) gives the largest impact score (0.872), while the velocity bias attack (Case 3) gives the smallest one (0.240). Among the vehicles, V4 is the most affected and V2 is the most resilient, with the lowest average impact score (0.362).

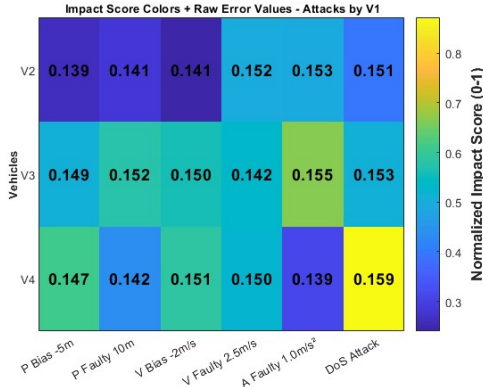


Fig. 2. Heatmap of normalized impact scores for all vehicles and attack cases.

Figure 3 shows the trust evolution for the DoS case. During $[10, 15]$ s, the trust scores of V2, V3, and V4 drop below the threshold $\theta_{\min} = 0.5$ and recover after the attack ends, showing that the proposed mechanism detects and isolates unreliable data online.

6. CONCLUSION AND FUTURE WORK

This paper presented a trust-aware distributed observer for resilient platoon state estimation under cyberattacks. The approach combines local and distributed observers with online trust evaluation to reduce the influence of unreliable data. Future work will couple trust more closely with control, extend trust evaluation to non-neighboring vehicles, and improve adaptation under dynamic communication topologies. We will also study learning-based trust estimation and matrix-gain observer designs.

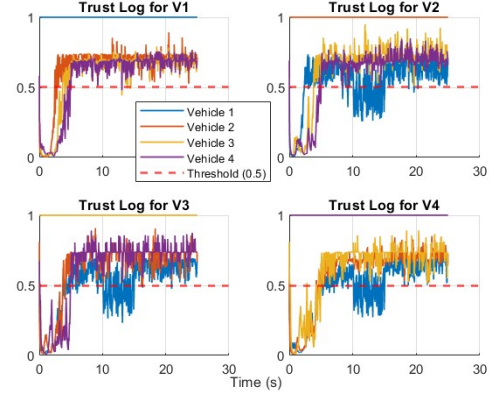


Fig. 3. Trust score evolution during DoS attack (Case 6).

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